Reconstruction Metric - GPU Optimized

2026-05-19

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0.1 1. Setup e Imports

```python
import numpy as np
import pandas as pd
import sys
import os
import subprocess
import time
import torch
import torch.nn as nn
import torch.nn.functional as F
import matplotlib.pyplot as plt

from pathlib import Path
from tqdm import tqdm
from sae.tools.experiment_utils import get_experiment_dir, get_metrics_dir, get_report_name
from sae.tools.naming_utils import get_checkpoint_dir, get_model_name, get_extras_id

project_root = Path('../../..').resolve()
sys.path.insert(0, str(project_root))

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f"Device: {device}")
if torch.cuda.is_available():
    print(f"GPU: {torch.cuda.get_device_name(0)}")
    print(f"Memoria: {torch.cuda.get_device_properties(0).total_memory / 1e9:.2f} GB")
```

```
Device: cuda
GPU: NVIDIA GeForce RTX 5060
Memoria: 8.55 GB
```

```python
# Configuración para visualización en PDF
pd.set_option('display.max_colwidth', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 100)
```

```python
np.set_printoptions(linewidth=100, edgeitems=3)

os.environ['COLUMNS'] = '100'

print(" Configuración para PDF lista")
```

```
 Configuración para PDF lista
```

```python
NAMING_META = {
    'variante': 'matryoshkabatchtopk',
    'tecnica':  'matryoshkabatchtopk',
    'capa':     6,
    'juegos':   25000,
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',
    'experiment_base_path': os.path.abspath('../../../experiments'),
    'extras': {
        'expansion':         16,
        'k':                 81,
        'matryoshka_widths': [2048, 4096, 8192],
        'epochs':            1000,
        'patience':          20,
        'lr':                1e-3
    }
}

print('\n Metadata de naming:')
for key, value in NAMING_META.items():
    print(f'  {key}: {value}')
```

```
 Metadata de naming:
  variante: matryoshkabatchtopk
  tecnica: matryoshkabatchtopk
  capa: 6
  juegos: 25000
  base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
  experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
  extras: {'expansion': 16, 'k': 81, 'matryoshka_widths': [2048, 4096, 8192], 'epochs': 1000, 'patience': 20, 'lr': 0.0
```

```python
class MatryoshkaBatchTopKSAE(nn.Module):
    """
    Sparse Autoencoder con activación MatryoshkaBatchTopK.
    Corregido según paper (ec. 3-5) e implementación de los autores (sae.py).

    Correcciones aplicadas:
    - P1: encoding único con W_enc completo; BatchTopK global; prefijos compartidos
    - P1: BatchTopK preserva valores originales (x * indicator), threshold EMA para eval
    - P1: auxiliary loss para dead features (sae.py:198-221)
    - P2: make_decoder_weights_and_grad_unit_norm proyecta gradiente (sae.py:51-57)
    - P2: preprocessing input_unit_norm configurable (sae.py:36-43)
    - P3: W_dec inicializado como transpuesta de W_enc (sae.py:93-94)
    """
    def __init__(self, d_in, d_sae, k, matryoshka_widths,
                 n_batches_to_dead=20, top_k_aux=512, aux_penalty=1/32,
                 input_unit_norm=False):
        super().__init__()
        self.d_in  = d_in
        self.d_sae = d_sae
        self.k     = k
        self.matryoshka_widths = sorted(matryoshka_widths)
        assert self.matryoshka_widths[-1] == d_sae, \
```

```python
            f"El último ancho ({self.matryoshka_widths[-1]}) debe ser d_sae ({d_sae})"

        self.n_batches_to_dead = n_batches_to_dead
        self.top_k_aux         = top_k_aux
        self.aux_penalty       = aux_penalty
        self.input_unit_norm   = input_unit_norm

        self.W_enc = nn.Parameter(torch.empty(d_in, d_sae))
        self.W_dec = nn.Parameter(torch.empty(d_sae, d_in))
        self.b_dec = nn.Parameter(torch.zeros(d_in))

        # P3: W_dec = W_enc^T normalizado (sae.py:93-94) - inicialización tied
        nn.init.kaiming_uniform_(self.W_enc)
        self.W_dec.data[:] = self.W_enc.t().data
        self.W_dec.data[:] = self.W_dec / self.W_dec.norm(dim=-1, keepdim=True)

        # P1: threshold persistente calibrado con EMA durante training (sae.py:97)
        self.register_buffer('threshold', torch.tensor(0.0))
        # P1: contador de batches sin activación por feature - para L_aux (sae.py:30-32)
        self.register_buffer('num_batches_not_active', torch.zeros(d_sae))

    # ------------------------------------------------------------------
    # P2: Preprocessing del input (sae.py:36-48)
    # ------------------------------------------------------------------

    def preprocess_input(self, x):
        if self.input_unit_norm:
            x_mean = x.mean(dim=-1, keepdim=True)
            x = x - x_mean
            x_std = x.std(dim=-1, keepdim=True)
            x = x / (x_std + 1e-5)
            return x, x_mean, x_std
        return x, None, None

    def postprocess_output(self, x_reconstruct, x_mean, x_std):
        if self.input_unit_norm and x_mean is not None:
            x_reconstruct = x_reconstruct * x_std + x_mean
        return x_reconstruct

    # ------------------------------------------------------------------
    # P1: Encoding único + BatchTopK global (sae.py:100-119)
    # ------------------------------------------------------------------

    def compute_activations(self, x_cent):
        # P1: encoding ÚNICO con W_enc completo - un solo pase para todos los anchos
        # Las features 0:w son literalmente las mismas en todos los niveles (paper ec. 3)
        pre_acts = x_cent @ self.W_enc     # [batch, d_sae]
        acts     = F.relu(pre_acts)         # ReLU antes del topk (sae.py:102)

        if self.training:
            # P1: BatchTopK global sobre d_sae×batch_size valores (paper ec. 6)
            # Preserva valores originales con scatter - NO resta threshold (sae.py:104-114)
            topk_result = torch.topk(acts.flatten(), self.k * x_cent.shape[0])
            acts_topk = (
                torch.zeros_like(acts.flatten())
                .scatter(-1, topk_result.indices, topk_result.values)
                .reshape(acts.shape)
            )
            self._update_threshold(acts_topk)
        else:
            acts.mul_(acts > self.threshold)
            return acts, acts
```

```python
        return acts, acts_topk

    @torch.no_grad()
    def _update_threshold(self, acts_topk, lr=0.01):
        # P1: EMA del mínimo valor positivo - calibra   para eval (sae.py:224-228)
        positive_mask = acts_topk > 0
        if positive_mask.any():
            min_positive = acts_topk[positive_mask].min()
            self.threshold = (1 - lr) * self.threshold + lr * min_positive

    @torch.no_grad()
    def _update_inactive_features(self, acts_topk):
        # P1: necesario para L_aux - trackea qué features no se activan (sae.py:60-62)
        self.num_batches_not_active += (acts_topk.sum(0) == 0).float()
        self.num_batches_not_active[acts_topk.sum(0) > 0] = 0

    # ------------------------------------------------------------------
    # Forward
    # ------------------------------------------------------------------

    def forward(self, x):
        """
        Retorna:
            outputs      : lista de (recon_w, hidden_w) para cada ancho matryoshka
                           recon_w está en espacio normalizado (= espacio original si input_unit_norm=False)
            all_acts     : pre-topk [batch, d_sae] - para L_aux
            all_acts_topk: post-topk [batch, d_sae]
            x_proc       : x preprocesado (= x si input_unit_norm=False)
        """
        x_proc, x_mean, x_std = self.preprocess_input(x)
        x_cent = x_proc - self.b_dec
        all_acts, all_acts_topk = self.compute_activations(x_cent)

        # P1: reconstrucciones por PREFIJO compartido (paper ec. 4, sae.py:149-155)
        # all_acts_topk[:, 0:w] son las mismas features en todos los niveles anidados
        # recon_w se mantiene en espacio normalizado para que la loss compare espacios iguales
        outputs = []
        for w in self.matryoshka_widths:
            hidden_w = all_acts_topk[:, :w]
            recon_w  = hidden_w @ self.W_dec[:w, :] + self.b_dec
            outputs.append((recon_w, hidden_w))

        self._update_inactive_features(all_acts_topk)
        return outputs, all_acts, all_acts_topk, x_proc

    # ------------------------------------------------------------------
    # P1: Auxiliary loss para dead features (sae.py:198-221)
    # ------------------------------------------------------------------

    def get_auxiliary_loss(self, x_proc, x_reconstruct_full, all_acts):
        residual        = x_proc.float() - x_reconstruct_full.float()
        aux_reconstruct = torch.zeros_like(residual)

        dead_features = self.num_batches_not_active >= self.n_batches_to_dead

        if dead_features.sum() > 0:
            acts_topk_aux = torch.topk(
                all_acts[:, dead_features],
                min(self.top_k_aux, int(dead_features.sum())),
                dim=-1,
            )
```

```
            acts_aux = torch.zeros_like(all_acts[:, dead_features]).scatter(
                -1, acts_topk_aux.indices, acts_topk_aux.values
            )
            aux_reconstruct = aux_reconstruct + acts_aux @ self.W_dec[dead_features]

        if aux_reconstruct.abs().sum() > 0:
            return self.aux_penalty * (aux_reconstruct.float() - residual.float()).pow(2).mean()
        return torch.tensor(0.0, device=x_proc.device)

    # ------------------------------------------------------------------
    # P2: Constraint de norma unitaria con proyección del gradiente (sae.py:51-57)
    # ------------------------------------------------------------------

    @torch.no_grad()
    def make_decoder_weights_and_grad_unit_norm(self):
        # P2: proyecta gradiente al espacio tangente de la esfera unitaria + normaliza
        # @torch.no_grad() evita trazar grafo interno - no impide leer .grad (sae.py:50)
        # Llamar ANTES de optimizer.step() (training.py:31)
        W_dec_normed    = self.W_dec / self.W_dec.norm(dim=-1, keepdim=True)
        W_dec_grad_proj = (
            (self.W_dec.grad * W_dec_normed).sum(-1, keepdim=True) * W_dec_normed
        )
        self.W_dec.grad -= W_dec_grad_proj
        self.W_dec.data  = W_dec_normed
```

0.2 2. Cargar Datos

```
# Cargar activaciones
activations_path = project_root / "activations" / "data" / "layer6_2000games.npy"
activations = np.load(activations_path)

print(f"Activaciones del modelo:")
print(f"  Shape: {activations.shape}")
print(f"  Memoria: {activations.nbytes / (1024**2):.2f} MB")
```

```
Activaciones del modelo:
  Shape: (118000, 512)
  Memoria: 230.47 MB
```

```
# Cargar ground truth de BSPs
bsp_gt_path = project_root /"metrics" / "02_data" / "bsp_ground_truth_2000games.npy"
bsp_names_path = project_root /"metrics" / "02_data" / "bsp_ground_truth_2000games.names.npy"

bsp_ground_truth = np.load(bsp_gt_path)
bsp_names = np.load(bsp_names_path, allow_pickle=True)

print(f"\nGround truth BSPs:")
print(f"  Shape: {bsp_ground_truth.shape}")
print(f"  Total BSPs: {len(bsp_names)}")
```

```
Ground truth BSPs:
  Shape: (118000, 326)
  Total BSPs: 326
```

0.3 3. Cargar SAE y Extraer Features

```python
# Configuración del SAE
input_dim = 512
hidden_dim = 8192

# Construir ruta del modelo usando naming_utils
checkpoint_dir = get_checkpoint_dir(
    NAMING_META['base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos']
)
extras_id = get_extras_id(checkpoint_dir, NAMING_META['extras']) if NAMING_META.get('extras') else None

model_filename = get_model_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'final',
    extras_id=extras_id
)
model_path = checkpoint_dir / model_filename

# Cargar modelo Matryoska SAE
sae = MatryoshkaBatchTopKSAE(d_in=input_dim, d_sae=hidden_dim, k=NAMING_META['extras']['k'], matryoshka_widths=NAMING_M

checkpoint = torch.load(model_path, map_location=device, weights_only=False)
sae.load_state_dict(checkpoint['model_state_dict'])
sae.eval()

print(f"SAE cargado:")
print(f"  Path: {model_path}")
print(f"  Input: {input_dim}, Hidden: {hidden_dim}, k: {NAMING_META['extras']['k']}")
print(f"  Expansion: {hidden_dim/input_dim}x")
```

```
SAE cargado:
  Path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_matryoshkabatchtopk\sae_matryoshkabatchtopk
  Input: 512, Hidden: 8192, k: 81
  Expansion: 16.0x
```

```python
# Extraer features del SAE en GPU
print("Extrayendo features del SAE...")
activations_tensor = torch.from_numpy(activations).float().to(device)

with torch.no_grad():
    outputs, all_acts, all_acts_topk, x_proc = sae(activations_tensor)
    sae_features = all_acts_topk

# Liberar tensores intermedios que ya no se necesitan
del outputs, all_acts, all_acts_topk, x_proc, activations_tensor
torch.cuda.empty_cache()

print(f"\nFeatures SAE:")
print(f"  Shape: {sae_features.shape}")
print(f"  Device: {sae_features.device}")
print(f"  Sparsity: {(sae_features == 0).float().mean():.2%}")
#print(f"  Activaciones promedio: {(sae_features > 0).sum(dim=1).float().mean():.1f}")
```

```
Extrayendo features del SAE...
```

```
Features SAE:
  Shape: torch.Size([118000, 8192])
  Device: cuda:0
  Sparsity: 99.01%
```

0.4 4. Split Train/Test y Filtrar BSPs de Piezas

```python
# Split 50/50:
n_moves = 59
n_games = 2000  # número de partidas para prueba de métricas
train_games = n_games // 2
split_idx = train_games * n_moves

sae_features_train = sae_features[:split_idx]
sae_features_test  = sae_features[split_idx:]

print(f"Total partidas: {n_games}")
print(f"Train: {train_games} partidas ({split_idx} activaciones)")
print(f"Test:  {n_games - train_games} partidas ({len(activations) - split_idx} activaciones)")
print(f"Train set: {sae_features_train.shape}")
print(f"Test set:  {sae_features_test.shape}")
```

```
Total partidas: 2000
Train: 1000 partidas (59000 activaciones)
Test:  1000 partidas (59000 activaciones)
Train set: torch.Size([59000, 8192])
Test set:  torch.Size([59000, 8192])
```

```python
# ============================================
# CONFIGURACIÓN DE FILTRADO DE BSPs
# ============================================
USE_PLAYER   = False  # BSPs perspectiva del jugador (1 y 2)
USE_ABSOLUTE = True    # BSPs absolutas negro/blanco (B y W)
USE_STRATEGIC = False  # BSPs especiales (BSP_)

bsp_pieces_indices = []
bsp_pieces_names = []

for i, name in enumerate(bsp_names):
    include = False
    if USE_PLAYER and len(name) == 6 and name.startswith('BSP') and (name.endswith('1') or name.endswith('2')):
        include = True
    if USE_ABSOLUTE and (name.endswith('B') or name.endswith('W')):
        include = True
    if USE_STRATEGIC and name.startswith('BSP_'):
        include = True
    if include:
        bsp_pieces_indices.append(i)
        bsp_pieces_names.append(name)

bsp_pieces_indices = np.array(bsp_pieces_indices)

# Para reconstruction necesitas split train/test
bsp_gt_train_pieces = torch.from_numpy(bsp_ground_truth[:split_idx, bsp_pieces_indices]).bool().to(device)
bsp_gt_test_pieces  = torch.from_numpy(bsp_ground_truth[split_idx:, bsp_pieces_indices]).bool().to(device)

# Convertir a tensor en GPU
# Identificador del filtrado (para naming de archivos de salida)
parts = []
if USE_PLAYER: parts.append("player")
```

```python
if USE_ABSOLUTE: parts.append("absolute")
if USE_STRATEGIC: parts.append("strategic")
filter_suffix = "_".join(parts) if parts else "none"

print(f"BSPs seleccionadas para Reconstruction:")
print(f"  Player: {USE_PLAYER} | Absolute: {USE_ABSOLUTE} | Strategic: {USE_STRATEGIC}")
print(f"  Total: {len(bsp_pieces_indices)}")
print(f"  Train shape: {bsp_gt_train_pieces.shape}")
print(f"  Test shape:  {bsp_gt_test_pieces.shape}")
print(f"  Device: {bsp_gt_train_pieces.device}")
print(f"  Primeras 5: {', '.join(bsp_pieces_names[:5])}")
print(f"  Últimas 5: {', '.join(bsp_pieces_names[-5:])}")
print(f"  Filter suffix: {filter_suffix}")
```

```
BSPs seleccionadas para Reconstruction:
  Player: False | Absolute: True | Strategic: False
  Total: 128
  Train shape: torch.Size([59000, 128])
  Test shape:  torch.Size([59000, 128])
  Device: cuda:0
  Primeras 5: BSPA1B, BSPA1W, BSPA2B, BSPA2W, BSPA3B
  Últimas 5: BSPH6W, BSPH7B, BSPH7W, BSPH8B, BSPH8W
  Filter suffix: absolute
```

0.5 5. Funciones GPU-Optimizadas

```python
def fast_precision_score_gpu(y_true, y_pred, eps=1e-8):
    """
    Precisión vectorizada en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions, n_candidates) booleano

    Returns:
        precision_scores: Tensor (n_candidates,)
    """
    y_true_expanded = y_true.unsqueeze(1)

    tp = (y_true_expanded & y_pred).sum(dim=0).float()
    fp = (~y_true_expanded & y_pred).sum(dim=0).float()

    precision = tp / (tp + fp + eps)

    return precision


def fast_f1_score_gpu(y_true, y_pred, eps=1e-8):
    """
    F1 score vectorizado en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions,) booleano

    Returns:
        f1: Float
    """
    tp = (y_true & y_pred).sum().float()
    fp = (~y_true & y_pred).sum().float()
```

```
    fn = (y_true & ~y_pred).sum().float()

    precision = tp / (tp + fp + eps)
    recall = tp / (tp + fn + eps)

    f1 = 2 * (precision * recall) / (precision + recall + eps)

    return f1.item()

print(" Funciones GPU definidas")
```

```
 Funciones GPU definidas
```

0.6 6. Fase 1: Identificar Features de Alta Precisión (GPU)

Para cada BSP, encontrar features con precisión   0.95 en el train set.

```
def identify_high_precision_features_gpu(
    sae_features_train,
    bsp_gt_train,
    f_max,
    precision_threshold=0.95,
    significance_threshold=1,
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
    feature_batch_size=512,
    verbose=True
):
    """
    Para cada (threshold t, feature f, BSP b): determina si f es clasificador
    de alta precisión para b al threshold t.

    Criterios (igual que implementación oficial):
        - precision(f, b, t) >= precision_threshold   (default 0.95)
        - on_count(f, t)     >= significance_threshold (default 10)

    Args:
        sae_features_train:    Tensor (P, n_features) GPU
        bsp_gt_train:          Tensor (P, n_bsps) GPU, bool
        f_max:                 Tensor (n_features,) GPU - calculado FUERA para
                               garantizar identidad exacta con Fase 2
        precision_threshold:   float, default 0.95
        significance_threshold: int, mín. activaciones requeridas (default 10)
        thresholds:            lista de fracciones t   [0, 1)
        feature_batch_size:    int, features por mini-batch interno

    Returns:
        hp_mask_TFB:            BoolTensor (T, n_active, n_bsps)
        active_feature_indices: LongTensor (n_active,)
    """
    n_positions, n_features = sae_features_train.shape
    n_bsps = bsp_gt_train.shape[1]
    device = sae_features_train.device

    thresholds_T = torch.tensor(thresholds, device=device)
    n_thresholds = len(thresholds_T)

    # Features vivas (f_max > 0)
    active_feature_indices = (f_max > 0).nonzero(as_tuple=True)[0]
    n_active = len(active_feature_indices)

    if verbose:
```

```python
    print("=" * 60)
    print("FASE 1: IDENTIFICAR FEATURES DE ALTA PRECISIÓN")
    print("=" * 60)
    print(f"Precision threshold:     {precision_threshold}")
    print(f"Significance threshold:  >= {significance_threshold} activaciones")
    print(f"Features activas:        {n_active}/{n_features} ({n_active/n_features*100:.1f}%)")
    print(f"BSPs:                    {n_bsps}")
    print(f"Thresholds:              {thresholds}")
    print("=" * 60)

# Thresholds absolutos: thresh_TF[t, f] = t * f_max[f]
active_f_max  = f_max[active_feature_indices]           # (n_active,)
thresholds_TF = thresholds_T[:, None] * active_f_max   # (T, n_active)

# Acumuladores sobre todo el train set
on_count_TF  = torch.zeros(n_thresholds, n_active, device=device)
tp_count_TFB = torch.zeros(n_thresholds, n_active, n_bsps, device=device)

bsp_float_PB = bsp_gt_train.float()  # (P, B)

n_iters = (n_active + feature_batch_size - 1) // feature_batch_size

for fi in tqdm(range(n_iters), desc="Fase 1"):
    f_start = fi * feature_batch_size
    f_end   = min(f_start + feature_batch_size, n_active)

    global_idx = active_feature_indices[f_start:f_end]
    acts_PF    = sae_features_train[:, global_idx]     # (P, bs)
    thresh_TF  = thresholds_TF[:, f_start:f_end]      # (T, bs)

    # active_TFP[t, f, p] = True si acts[p,f] > thresh[t,f]
    active_TFP = acts_PF.T.unsqueeze(0) > thresh_TF.unsqueeze(2)  # (T, bs, P)

    # Conteo de activaciones por (threshold, feature)
    on_count_TF[:, f_start:f_end] += active_TFP.sum(dim=2).float()

    # True positives: f activa Y BSP verdadera
    # tp[t, f, b] = Σ_p active[t,f,p] * bsp[p,b]
    tp_count_TFB[:, f_start:f_end, :] += torch.einsum(
        'tfp,pb->tfb', active_TFP.float(), bsp_float_PB
    )

# Precisión: tp / on_count
eps = 1e-8
precision_TFB = tp_count_TFB / (on_count_TF.unsqueeze(2) + eps)  # (T, F, B)

# Máscara: alta precisión AND significancia suficiente
sig_mask_TF1 = (on_count_TF >= significance_threshold).unsqueeze(2)  # (T, F, 1)
hp_mask_TFB  = (precision_TFB >= precision_threshold) & sig_mask_TF1  # (T, F, B)

if verbose:
    hp_per_t = hp_mask_TFB.sum(dim=(1, 2))  # (T,)
    hp_per_b = hp_mask_TFB.sum(dim=1)        # (T, B)
    best_t   = hp_per_t.argmax().item()
    print(f"\nPares (feature, BSP) de alta precisión por threshold:")
    for i, t in enumerate(thresholds):
        marker = " ← más pares" if i == best_t else ""
        print(f"  t={t:.1f}: {hp_per_t[i].item():5d} pares{marker}")
    print(f"\nFeatures HP por BSP en t={thresholds[best_t]:.1f}:")
    print(f"  Media:             {hp_per_b[best_t].float().mean():.1f}")
    print(f"  Máx:               {hp_per_b[best_t].max().item()}")
    print(f"  BSPs sin features: {(hp_per_b[best_t] == 0).sum().item()}")
```

```
    return hp_mask_TFB, active_feature_indices

print("  Fase 1 rediseñada: umbral de significancia ( 10) + estructura (T, F, B)")
```

```
  Fase 1 rediseñada: umbral de significancia ( 10) + estructura (T, F, B)
```

```
THRESHOLDS = [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]

# f_max calculado UNA SOLA VEZ aquí - se pasa a Fase 1 y Fase 2
f_max = sae_features_train.max(dim=0)[0]  # (n_features,)
print(f"f_max calculado: {(f_max > 0).sum().item()} features activas de {f_max.shape[0]}")

start_time = time.time()

hp_mask_TFB, active_feature_indices = identify_high_precision_features_gpu(
    sae_features_train,
    bsp_gt_train_pieces,
    f_max,
    precision_threshold=0.95,
    significance_threshold=1,
    thresholds=THRESHOLDS,
    feature_batch_size=512,
    verbose=True
)

phase1_time = time.time() - start_time

print(f"\nTiempo Fase 1: {phase1_time:.2f} seg")
print(f"hp_mask_TFB shape: {hp_mask_TFB.shape}  (T, n_active, n_bsps)")
```

```
f_max calculado: 8191 features activas de 8192
============================================================
FASE 1: IDENTIFICAR FEATURES DE ALTA PRECISIÓN
============================================================
Precision threshold:     0.95
Significance threshold:  >= 1 activaciones
Features activas:        8191/8192 (100.0%)
BSPs:                    128
Thresholds:              [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]
============================================================

Fase 1: 100%|     | 16/16 [00:00<00:00, 1656.44it/s]


Pares (feature, BSP) de alta precisión por threshold:
  t=0.0:   631 pares
  t=0.1:   643 pares
  t=0.2:  1045 pares
  t=0.3:  2908 pares
  t=0.4:  5096 pares
  t=0.5:  7547 pares
  t=0.6: 10873 pares
  t=0.7: 18930 pares
  t=0.8: 44457 pares
  t=0.9: 109852 pares ← más pares

Features HP por BSP en t=0.9:
  Media:             858.2
  Máx:               2997
  BSPs sin features: 0
```

```
Tiempo Fase 1: 0.88 seg
hp_mask_TFB shape: torch.Size([10, 8191, 128])  (T, n_active, n_bsps)
```

0.7 7. Fase 2: Reconstruir Tableros en Test Set (GPU)

```python
def reconstruct_boards_gpu(
    sae_features_test,
    bsp_gt_test,
    hp_mask_TFB,
    active_feature_indices,
    f_max,
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
    verbose=True
):
    """
    Fase 2: barrido simultáneo de thresholds T → toma el mejor F1.

    Para cada threshold t:
      1. Binarizar features de test: active[p, f] = acts[p, f] > t * f_max[f]
      2. Predicción por (posición, BSP):
             predict[p, b] = OR_f ( active[p,f] AND hp_mask[t, f, b] )
             implementado como: (active_PF @ hp_FB) > 0
      3. F1 global micro-averaged (acumula TP/FP/FN sobre todas las posiciones y BSPs)

    Retorna el max F1 sobre todos los thresholds (Eq. 5 del paper: max_t).

    Args:
        sae_features_test:      Tensor (P, n_features) GPU
        bsp_gt_test:            Tensor (P, n_bsps) GPU, bool
        hp_mask_TFB:            BoolTensor (T, n_active, n_bsps) de Fase 1
        active_feature_indices: LongTensor (n_active,) de Fase 1
        f_max:                  Tensor (n_features,) GPU - MISMO objeto que Fase 1
        thresholds:             lista de fracciones (debe coincidir con Fase 1)

    Returns:
        best_f1:          float, max F1 sobre thresholds
        best_threshold:   float, threshold óptimo
        f1_per_threshold: np.array (T,), F1 por threshold
    """
    device = sae_features_test.device
    n_thresholds = len(thresholds)
    thresholds_T = torch.tensor(thresholds, device=device)

    # Activaciones del test solo para features activas
    test_active_PF = sae_features_test[:, active_feature_indices]  # (P, n_active)
    active_f_max   = f_max[active_feature_indices]                  # (n_active,)

    gt_PB = bsp_gt_test.bool()  # (P, B)
    eps   = 1e-8

    f1_scores   = torch.zeros(n_thresholds, device=device)
    precision_T = torch.zeros(n_thresholds, device=device)
    recall_T    = torch.zeros(n_thresholds, device=device)

    if verbose:
        print("=" * 60)
        print("FASE 2: RECONSTRUCCIÓN - BARRIDO DE THRESHOLDS")
        print("=" * 60)
        print(f"{'t':>5}  {'F1':>8}  {'P':>8}  {'R':>8}  {'TP':>8}  {'FP':>8}  {'FN':>8}")
        print("-" * 60)
```

```python
    for t_idx in range(n_thresholds):
        t = thresholds[t_idx]

        # Threshold absoluto por feature: t * f_max[f]
        abs_thresh_F = thresholds_T[t_idx] * active_f_max  # (n_active,)

        # Binarizar features en test
        active_PF = test_active_PF > abs_thresh_F.unsqueeze(0)  # (P, n_active)

        # predict[p, b] = any( active[p,f] AND hp[t,f,b] )
        # = (active_PF.float() @ hp_mask[t].float()) > 0
        hp_FB          = hp_mask_TFB[t_idx].float()              # (n_active, B)
        predictions_PB = (active_PF.float() @ hp_FB) > 0         # (P, B)

        # Métricas globales micro-averaged
        tp = (predictions_PB &  gt_PB).sum().float()
        fp = (predictions_PB & ~gt_PB).sum().float()
        fn = (~predictions_PB & gt_PB).sum().float()

        p  = tp / (tp + fp + eps)
        r  = tp / (tp + fn + eps)
        f1 = 2 * p * r / (p + r + eps)

        f1_scores[t_idx]   = f1
        precision_T[t_idx] = p
        recall_T[t_idx]    = r

        if verbose:
            print(f"{t:>5.1f}  {f1.item():>8.4f}  {p.item():>8.4f}  {r.item():>8.4f}"
                  f"  {tp.item():>8.0f}  {fp.item():>8.0f}  {fn.item():>8.0f}")

    best_idx       = f1_scores.argmax().item()
    best_f1        = f1_scores[best_idx].item()
    best_threshold = thresholds[best_idx]

    if verbose:
        print("=" * 60)
        print(f"Mejor threshold: t={best_threshold:.1f}")
        print(f"Reconstruction Score (max_t F1): {best_f1:.4f}")
        print("=" * 60)

    return best_f1, best_threshold, f1_scores.cpu().numpy()

print("  Fase 2 rediseñada: barrido T simultáneo + F1 global micro-averaged + max_t")
```

```
  Fase 2 rediseñada: barrido T simultáneo + F1 global micro-averaged + max_t
```

```python
start_time = time.time()

reconstruction_score, best_threshold, f1_per_threshold = reconstruct_boards_gpu(
    sae_features_test,
    bsp_gt_test_pieces,
    hp_mask_TFB,
    active_feature_indices,
    f_max,           # mismo tensor calculado antes de Fase 1
    thresholds=THRESHOLDS,
    verbose=True
)

phase2_time = time.time() - start_time
total_time  = phase1_time + phase2_time
```

```
print(f"\nTiempo Fase 2: {phase2_time:.2f} seg")
print(f"Tiempo total:  {total_time:.2f} seg ({total_time/60:.2f} min)")
```

```
============================================================
FASE 2: RECONSTRUCCIÓN - BARRIDO DE THRESHOLDS
============================================================
    t        F1         P         R        TP        FP        FN
------------------------------------------------------------
  0.0    0.0112    0.9402    0.0056     11317       720   1994683
  0.1    0.0137    0.9437    0.0069     13867       828   1992133
  0.2    0.0227    0.9429    0.0115     23006      1394   1982994
  0.3    0.0364    0.9403    0.0185     37185      2359   1968815
  0.4    0.0431    0.9359    0.0221     44285      3032   1961715
  0.5    0.0470    0.9215    0.0241     48332      4117   1957668
  0.6    0.0491    0.8853    0.0252     50641      6559   1955359
  0.7    0.0529    0.7809    0.0274     54918     15409   1951082
  0.8    0.0661    0.6204    0.0349     70006     42839   1935994
  0.9    0.0872    0.5042    0.0477     95754     94159   1910246
============================================================
Mejor threshold: t=0.9
Reconstruction Score (max_t F1): 0.0872
============================================================

Tiempo Fase 2: 11.79 seg
Tiempo total:  12.67 seg (0.21 min)
```

0.8 8. Análisis de Resultados

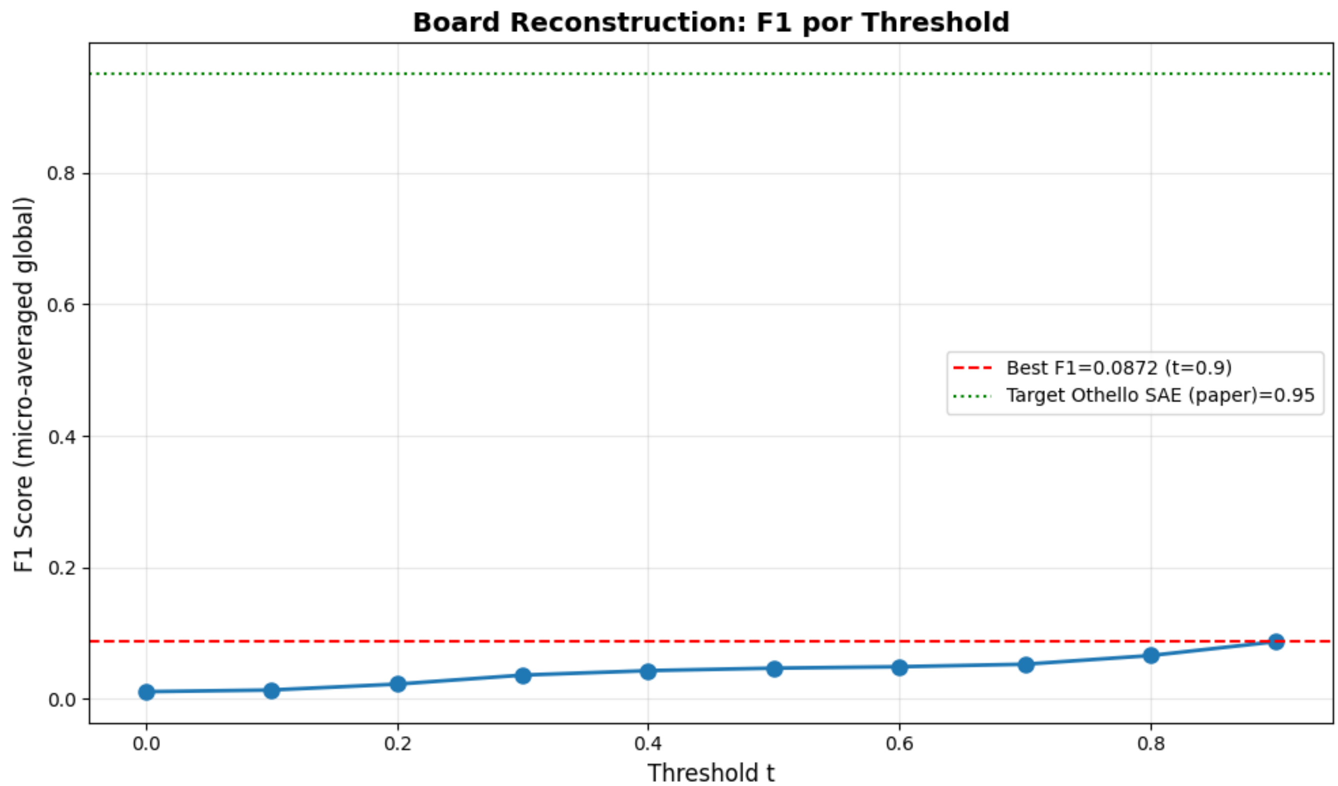
```
print("=" * 60)
print("BARRIDO DE THRESHOLDS - F1 POR THRESHOLD")
print("=" * 60)
for i, (t, f1) in enumerate(zip(THRESHOLDS, f1_per_threshold)):
    marker = " ← MEJOR" if i == f1_per_threshold.argmax() else ""
    print(f"t={t:.1f}: F1={f1:.4f}{marker}")
print("=" * 60)

plt.figure(figsize=(10, 6))
plt.plot(THRESHOLDS, f1_per_threshold, marker='o', linewidth=2, markersize=8)
plt.axhline(y=reconstruction_score, color='r', linestyle='--',
            label=f'Best F1={reconstruction_score:.4f} (t={best_threshold:.1f})')
plt.axhline(y=0.95, color='g', linestyle=':', label='Target Othello SAE (paper)=0.95')
plt.xlabel('Threshold t', fontsize=12)
plt.ylabel('F1 Score (micro-averaged global)', fontsize=12)
plt.title('Board Reconstruction: F1 por Threshold', fontsize=14, fontweight='bold')
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()

print(f"\n  Threshold óptimo: t={best_threshold:.1f}")
print(f"  Reconstruction Score: {reconstruction_score:.4f}")
```

```
============================================================
BARRIDO DE THRESHOLDS - F1 POR THRESHOLD
============================================================
t=0.0: F1=0.0112
t=0.1: F1=0.0137
t=0.2: F1=0.0227
t=0.3: F1=0.0364
```

```
t=0.4: F1=0.0431
t=0.5: F1=0.0470
t=0.6: F1=0.0491
t=0.7: F1=0.0529
t=0.8: F1=0.0661
t=0.9: F1=0.0872 ← MEJOR
============================================================
```




```
  Threshold óptimo: t=0.9
  Reconstruction Score: 0.0872
```

```
# Estadísticas sobre features de alta precisión por BSP
hp_per_bsp = hp_mask_TFB.sum(dim=1)  # (T, n_bsps)
best_t_idx = torch.tensor(f1_per_threshold).argmax().item()
n_features_per_bsp = hp_per_bsp[best_t_idx].cpu().numpy()

print("="*60)
print("ESTADÍSTICAS - FEATURES POR BSP")
print("="*60)
print(f"Total BSPs: {len(n_features_per_bsp)}")
print(f"Features promedio por BSP: {np.mean(n_features_per_bsp):.1f}")
print(f"Features mediana por BSP: {np.median(n_features_per_bsp):.1f}")
print(f"BSPs con 0 features: {np.sum(n_features_per_bsp == 0)}")
print(f"BSPs con 1+ features: {np.sum(n_features_per_bsp > 0)}")
print(f"BSPs con 5+ features: {np.sum(n_features_per_bsp >= 5)}")
print(f"BSPs con 10+ features: {np.sum(n_features_per_bsp >= 10)}")
print(f"Max features en una BSP: {np.max(n_features_per_bsp)}")
print("="*60)
```

```
============================================================
ESTADÍSTICAS - FEATURES POR BSP
============================================================
```

```
Total BSPs: 128
Features promedio por BSP: 858.2
Features mediana por BSP: 690.0
BSPs con 0 features: 0
BSPs con 1+ features: 128
BSPs con 5+ features: 128
BSPs con 10+ features: 128
Max features en una BSP: 2997
============================================================
```

0.9 9. Guardar Resultados

```python
results = {
    'reconstruction_score': reconstruction_score,
    'best_threshold': best_threshold,
    'f1_per_threshold': f1_per_threshold,
    'thresholds': np.array(THRESHOLDS),
    'phase1_time_seconds': phase1_time,
    'phase2_time_seconds': phase2_time,
    'total_time_seconds': total_time,
    'bsp_names': bsp_pieces_names,
}

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)
output_path = metrics_dir / f"reconstruction_results_{filter_suffix}.npz"
np.savez(output_path, **results)

print(f"  Resultados guardados en: {output_path}")
print(f"\nContenido:")
print(f"  reconstruction_score: {reconstruction_score:.4f}")
print(f"  best_threshold:       {best_threshold:.1f}")
print(f"  f1_per_threshold:     {f1_per_threshold}")

# Comparación con el paper
print()
print("=" * 60)
print("COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)")
print("=" * 60)
print(f"{'Modelo':<25} {'Reconstruction':>14}")
print("-" * 40)
print(f"{'SAE trained (paper)':<25} {'0.95':>14}  ← Objetivo Othello")
print(f"{'Nuestro SAE':<25} {reconstruction_score:>14.4f}  ← Resultado")
print("=" * 60)

if reconstruction_score >= 0.90:
    print("\n  Excelente: Muy cercano al objetivo del paper")
elif reconstruction_score >= 0.50:
    print("\n~ Moderado: Por encima de random, revisar SAE")
else:
    print("\n  Bajo: Revisar entrenamiento del SAE o calidad de las activaciones")
```

```
 Resultados guardados en: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_matryoshkabatchto
```

```
Contenido:
  reconstruction_score: 0.0872
  best_threshold:       0.9
  f1_per_threshold:     [0.01121585 0.01372498 0.02266154 0.03635708 0.04313508 0.04695963 0.04908976 0.05289918 0.066
 0.08721111]

============================================================
COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)
============================================================
Modelo                    Reconstruction
----------------------------------------
SAE trained (paper)                 0.95  ← Objetivo Othello
Nuestro SAE                       0.0872  ← Resultado
============================================================

 Bajo: Revisar entrenamiento del SAE o calidad de las activaciones
```

0.10 10. Generar PDF con Quarto

```python
output_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    f'reconstruction_{filter_suffix}',
    extras_id=extras_id
)

notebook_name = 'reconstruction_sae_matryoska.ipynb'
output_path = output_dir / pdf_name

print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{output_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)
```

```
 Generando PDF con Quarto...
 Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_matryoshkabatchtopk\sae_
 PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_matryoshkabatcht
```